Journal of Public Transportation 28 (2026) 100146

Contents lists available at ScienceDirect

Journal of Public Transportation

journal homepage: www.elsevier.com/locate/jpubtr

Probabilistic modeling of delays for train journeys with transfers

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ARTICLE INFO

Keywords:
Railway
Train delay prediction
Reliability
Bayesian regression
Gradient boosting

ABSTRACT

Reliability plays a key role in the experience of a rail traveler. The reliability of journeys involving transfers is affected by the reliability of the transfers and the consequences of missing a transfer, as well as the possible delay of the train used to reach the destination. In this paper, we propose a flexible method to model the reliability of train journeys with any number of transfers. The method combines a transfer reliability model based on gradient boosting, responsible for predicting the reliability of transfers between trains, and a delay model based on probabilistic Bayesian regression, which is used to model train arrival delays. The models are trained on delay data from four Swedish train stations and evaluated on delay data from another two stations, in order to evaluate the generalization performance of the models. We show that the probabilistic delay model, which models train delays following a mixture distribution with two lognormal components, allows to much more realistically model the distribution of actual train delays compared to a standard lognormal model. Finally, we show how these models can be used together to predict the arrival delay distribution at the final destination of the journey. The results indicate that the method accurately predicts the reliability for nine out of ten tested journeys. The method could be used to improve journey planners by providing reliability information to travelers. Further applications include timetable planning and transport modeling.

1. Introduction

Reliability plays a key role in the experience of a rail traveler. The willingness of travelers to travel by train decreases when reliability is expected to be low (van Loon et al., 2011; Rietveld et al., 2001). For journeys involving more than one train, travelers face additional uncertainty due to the risk of missing a transfer and the consequences for the onward journey (Yap et al., 2024; Chowdhury and Ceder, 2016). Despite the importance of reliability for travelers, journey planning services usually return itineraries according to the schedule without any information on the reliability to be expected for a particular journey. A better understanding of the reliability of whole journeys is also a key to inform planners and railway operators to be able to make decisions improving the reliability perceived by travelers.

In this paper, we characterize the reliability of a journey using the distribution of the arrival delay at the final destination. The delay distribution gives detailed information about the deviations from the scheduled arrival at the final destination that are expected for a given journey. Three main factors impact arrival delay:

1. the probability that a transfer between trains during the journey is reached,
2. the consequence in case of missing a transfer, and
3. the arrival delay of the last actual train used to reach the destination.

In this paper, we aim to predict the arrival delay distribution for a given train journey with transfers, giving the probability and magnitude of deviations compared to the scheduled arrival. The purpose is to predict the typical reliability of a journey, that is, before any real-time information on specific delays is known. We use historic railway traffic data from Sweden to train and validate the method.

The main contribution of this paper is a flexible method for estimating the reliability of train journeys, including those with one or more transfers. As part of this method, we propose a model using gradient boosting to estimate the transfer reliability, that is, the probability that a transfer between two trains is reached, taking into account whether a previous transfer has been missed. We propose a probabilistic model of train arrival delays using a mixture lognormal distribution that outperforms a standard lognormal model. The proposed method combines both models to predict the reliability of a given train journey. The main benefit of the chosen model structure is its ability to generate probability distributions of the arrival delay, instead of just a point estimate. Since predictions are made for future journeys, the uncertainty of a point estimate of the delay would be quite high. Generating a distribution of the delay allows us to provide more useful results and

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https://doi.org/10.1016/j.jpubtr.2025.100146
Received 11 March 2025; Received in revised form 12 September 2025; Accepted 12 December 2025
Available online 19 December 2025
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metrics to the user. We validate the predictions using historic train traffic data from several train stations in Sweden.

The proposed method could provide reliability information in a journey planner alongside the scheduled itinerary. While commuters are often already acquainted with the particularities of their daily journey, this would also allow also infrequent travelers to take reliability into account when planning their journey, improving their travel experience. In order to decide which connections to sell, railway operators often use fixed minimum transfer time requirements, not taking the consequences of missing a particular connection for the onward journey into account. The proposed method could be used by operators to identify unreliable journeys and offer only journeys with a certain reliability.

The remainder of the paper is organized as follows. In Section 2, we review related work. Section 3 describes the dataset, including its sources, structure, and pre-processing steps. In Section 4, we outline the proposed method, detailing the models we developed. Section 5 presents the results, highlighting key findings and their implications. In Section 6, we discuss the results, their implications, and address the limitations of the method. Finally, Section 7 concludes the paper by summarizing the main contributions and suggesting directions for future research.

2. Related work

The concept of travel time reliability in transportation has been studied extensively (Taylor, 2013). Several studies used statistical models to describe the travel time variability in road networks (Fosgerau and Fukuda, 2012; Ma et al., 2017; Susilawati et al., 2013) and road-based public transport (Büchel and Corman, 2020). While these studies focus on describing the travel time reliability as an important feature of the performance of transportation networks, travel behavior researchers have studied how individual travelers account for reliability in their travel plans (Noland and Polak, 2002). Both stated preference (Rietveld et al., 2001; Asensio and Matas, 2008) and revealed preference (van Loon et al., 2011) studies indicate that the reliability of public transportation systems affects travelers' choices.

Within rail traffic, extensive research has been conducted on the prediction and modeling of train delays (Tiong et al., 2023). These approaches can generally be divided into long-term versus short-term predictions and event-driven versus data-driven predictions (Spanninger et al., 2022). Short-term predictions involve real-time data and are used to inform travelers who are currently traveling. Long-term predictions without using real-time data can be used to provide information during the route planning and ticket booking stages. Event-driven approaches incorporate a dependency structure between train events (arrivals and departures) and construct a network of consecutive events. Graph-based models such as Bayesian networks (Corman and Kecman, 2018) have been used to model this problem among others. Data-driven approaches, on the other hand, have shown increasing interest in recent years (Tiong et al., 2023). These methods predict the train delay directly, without modeling a dependency structure or intermediate predictions. Probability distributions like the exponential, lognormal, or Weibull distributions can be used to model train delays (Marković et al., 2015). Data-driven approaches often use machine learning methods such as neural networks (Luo et al., 2023) or decision tree-based methods (Shi et al., 2021).

While the previously mentioned approaches show good performance for predicting train delays at a certain station or for a single leg, they do not consider complete journeys with transfers between trains. Bates et al. (2001) reviewed the concepts of reliability and travel time variability for public transport journeys with transfers from a theoretical perspective. Few studies have modeled the reliability of journeys with transfers using real-world data. Keyhani et al. (2017) proposed a method that computes the reliability of train journeys with transfers using a probabilistic graph model based on empirical observations of train delays. Stratil-Sauer (2024) proposed a framework for modeling the arrival delay of train journeys with one transfer using a probabilistic Bayesian regression model. The problem of finding a reliable and robust route for a given journey has been studied by Böhmová et al. (2013) and Müller-Hannemann and Schnee (2004) among others. Liu et al. (2020) investigated the connectivity reliability of urban transport networks, identifying important interchange stations and causes of delays in the network.

The reliability of a journey can be described in different ways. Bates et al. (2001) uses the arrival time distribution to illustrate the probability of different delays at the destination. While this is a comprehensive way to describe the reliability of a single journey, it is difficult to communicate to travelers and hard to compare between journeys. A simpler reliability metric called *reliability rating* was introduced by Keyhani et al. (2012). It specifies in percent how many cases a traveler can expect to reach all transfers on a journey and arrive at the destination with the train they intended to take. Dixit et al. (2019) extends the *reliability buffer time* metric first introduced by Chan (2007) and Uniman et al. (2010) to journeys with multiple legs. This metric measures the variability in travel time a traveler can expect. It is defined as the difference between an extreme nth percentile (usually set to 95) and the 50th percentile travel time and can be interpreted as the additional time passengers have to account for during their journey to ensure arrival during this time period 19 out of 20 times.

3. Data

The dataset used in this paper consists of train arrival and departure information of passenger trains from multiple train stations in Sweden. The data were collected using an API provided by the Swedish Transport Agency, Trafikverket[1] between December 2022 and August 2024. The stations of Alvesta, Norrköping, Karlskrona, and Katrineholm are used to train the models. Observations from the stations of Hässleholm and Borlänge form the external test dataset used to judge the model performance on stations on which it was not trained. This is important to validate the generalization performance of the model (Tiong et al., 2023). Fig. 1 shows the location of the selected stations in the Swedish rail network. Norrköping, Alvesta, and Hässleholm are important hubs for trains on Sweden's southern mainline connecting Malmö and Stockholm, and many smaller connecting regional lines. Katrineholm is included because it lies on the western mainline that connects Stockholm and Göteborg. Karlskrona and Borlänge are included in the training and test data sets, respectively, because they are located off the main train lines and thus provide train events with different characteristics than the other stations.

Each observation in the dataset denotes a train stopping at a station and contains the following information:

- Train metadata (operator, origin, destination, train type)
- Scheduled time of arrival and departure
- Actual time of arrival and departure

Trains are separated into two types: regional trains and intercity (long-distance) trains. Since we focus on journeys that can be completed on a single day, night trains are excluded, as well as special trains such as museum trains. Intercity trains are all trains denoted *IC* or *Snabbtåg* by the state-owned operator SJ and all trains by the open-access operators Snälltåget and VR. All other trains are regional trains.

The observations only contain the events for a specific station. To obtain additional information about the train's route, these observations are joined with timetable data following the General Transit Feed Specification (GTFS).[2] We use these data to obtain each train's *runtime*

[1] https://data.trafikverket.se/

[2] https://www.trafiklab.se/api/gtfs-datasets/gtfs-sweden/

Fig. 1. Location of the stations used in model training and validation with the Swedish railway network according to jnbkarta.trafikverket.se in the background.

from its origin to the current station and also the *total runtime* between the origin and the final destination.

In a few cases, this join is unsuccessful, and no or faulty runtime information was obtained and therefore, these observations are discarded. Observations with missing or faulty delay information are filtered out as well. In total, this filtering process removes 9% of the observations. 159 266 observations remain that are distributed as follows between the stations: 59 356 in Alvesta, 6 159 in Karlskrona, 33 056 in Katrineholm, and 60 095 in Norrköping. Of these observations, we randomly select 80% as training data (127 302 observations) and the remaining 20% as the regular test dataset (31 964 observations). The external test dataset consists of 6 000 observations in Hässlehom and 6 000 in Borlänge. The models are evaluated on both the regular and the external test datasets.

3.1. Transfer data

Each transfer consists of an arrival event for the incoming train and a departure event for a train to which the traveler may transfer. All trains scheduled to leave the station between 3 and 60 min after an incoming train is scheduled to arrive are considered a transfer. This leaves us with 692 191 transfers in the training dataset and 173 421 in the test dataset derived from the train arrival and departure data. The external test dataset consists of 13 309 observations.

The *PlannedTransferTime* (PTT) for a transfer is defined as the time between the arrival of the incoming train and the departure of the connecting train, according to the schedule. A transfer is considered *Reached* if there are at least 3 min between the actual arrival of the incoming train and the actual departure of the connecting train. This limit is set to account for the time it takes to alight the train and walk to another platform.

3.2. Exploratory data analysis

Train delays of passenger trains generally show a right-skewed distribution. Although some trains arrive slightly early, many more trains arrive delayed, and the delays can occasionally be very large. Fig. 2 and Table 1 show that the maximum delay in our dataset is more than 7 h, while the mean delay is 4 min. The right-skewed nature of the distribution is confirmed by the fact that the mean is higher than the median and the positive skewness of 6.996.

Table 1
Summary statistics of the arrival delay (in minutes).

Min	Median	Mean	Max	Var.	Skewness	Kurtosis
−5.98333	0.5333	4.117	376.267	188.209	6.996	79.2104

Key patterns and interactions between the target variables (arrival delay and transfer reliability) and the available features are outlined in the following paragraphs.

Fig. 3 shows that there is a correlation between the features time of the day and train type and the arrival delay. Intercity trains have higher delays in general, and the average delay for these trains increases throughout the day, while the impact of the time of the day on the delay of regional trains is much smaller. This indicates an interaction effect between these features.

Fig. 4 shows that the reliability of transfers increases drastically when the available transfer time is longer. In total, 94.8% of the transfers in the dataset are reached and only 5.2% are missed. However, there is significant variation depending on, for example, the type of arriving train (regional or intercity) and the PTT. For short transfers with a PTT of at most 10 min, the transfer reliability drops to 80%, while transfers with a PTT of more than 30 min show an average transfer reliability as high as 98%. The reliability of transfers is naturally connected to the delay of a train, and the predictors mentioned above are also relevant in this context.

4. Method

In this section, we propose a method to estimate the reliability of a given train journey, described by a planned itinerary, including one or more transfers. The method uses two models as building blocks: A transfer reliability model that is responsible for predicting the reliability of a transfer, and a delay model, which is used to model train arrival delays. Both models are trained using historical train information. Finally, we present how both models can be used together to sample the arrival delay at the final destination of the journey, describing the reliability of the journey.

Let J be a train journey consisting of one or more legs. Legs are denoted as J^i with $\overline{J}$ specifying the total number of legs in the journey. A journey also has a scheduled arrival time at the final station as $t_s(J)$. The arrival delay of a leg is denoted as $Delay(J^i)$. Two consecutive legs of a journey are connected by a transfer, which is denoted as $Tr(J^i, J^{i+1}) \in \{0, 1\}$. If the transfer was missed, we set $Tr = 0$ and if it was reached $Tr = 1$. Table 2 gives an overview of the notations used throughout the rest of the paper.

4.1. Transfer model

The *transfer model* aims to predict the probability $Pr(Tr)$ of reaching a particular transfer Tr. It also gives the probability $Pr(Tr|Tr_{prev})$ of reaching a transfer Tr, conditional on an earlier, preferred transfer Tr_{prev} being missed.

To model the probability of reaching a transfer, we use an XGBoost model (Chen, 2015). XGBoost relies on gradient boosting of decision trees to predict the outcome of regression or classification problems. Predicting the reliability of transfers is a binary classification task, with imbalanced class proportions, since only about 5% of the transfers in the training data are missed. Decision Tree based methods and XGBoost in particular are well suited for this problem because they can easily handle NA values, work well with imbalanced data, and show

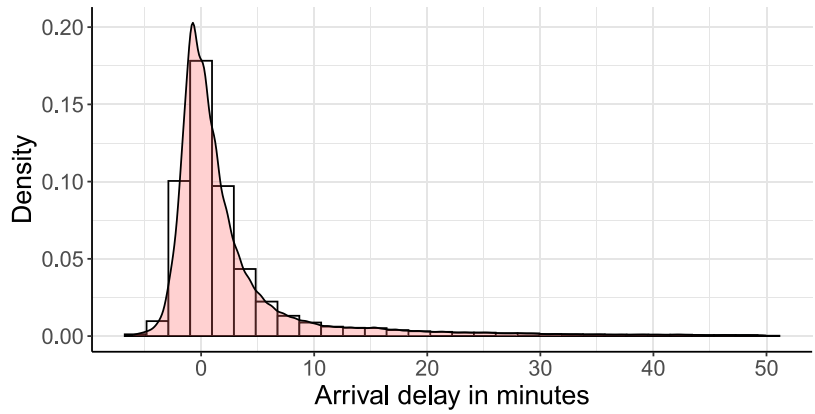

Fig. 2. Distribution of arrival delays in minutes. Total sample size of 159 266 observations, including 2 787 observations with a delay higher than 50 min that were removed for better readability.

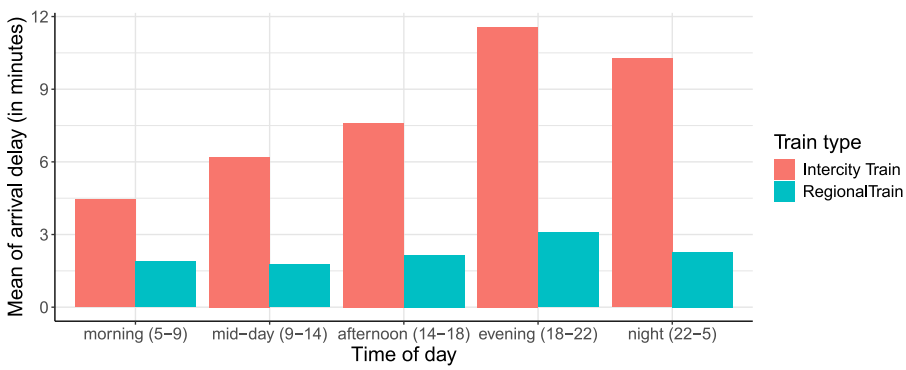

Fig. 3. Train arrival delay per train type and time of day.

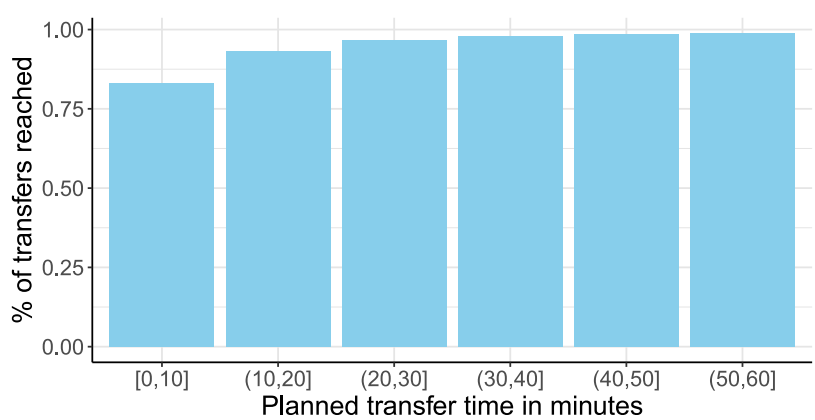

Fig. 4. Transfer reliability depending on the PTT.

good performance in classification tasks when compared with other methods (Khan et al., 2024).

An XGBoost model has several hyperparameters, including the number of trees to fit, the depth of these trees, and the size of the sample used to fit each iteration, among others. The right hyperparameters depend on the given data and have to balance between optimal prediction performance and overfitting concerns. To find the optimal setup, we use grid search using cross-validation on training data.

Features. Table 3 presents the features used in the transfer model. It contains features about the transfer, as well as characteristics of the arriving and departing leg. Some features are combined interaction variables, motivated by the relationships found in Section 3.2. For example, in Fig. 3, the delay of intercity trains depends strongly on the time of the day, while regional trains do not show this behavior. To reflect this interaction, the feature *arr. IntercityHour* is included. A similar interaction is the reason behind the *arr. IntercityWinter* feature. While regional trains did not show varying delays throughout the seasons, intercity trains were more often delayed in winter. The *Weekend* feature is a proxy for the exact weekday on which the journey takes place. Exploratory data analysis revealed no significant differences in transfer reliability or delays between Saturday and Sunday. But a difference between weekends and other days of the week could be shown.

The feature $arr.ShortTrain$ is used instead of including the runtime directly as a feature because this led to overfitting issues, where the model would interpret it as categorical instead of continuous, resulting in unrealistic predictions. For example, for a sample with a runtime of

Table 2
Overview of the notations used in the description of the method.

Notation	Definition
J	A journey
$\overline{J}$	The number of legs in the journey
J^i	The ith leg of journey J, $i \in \{1, \dots, \overline{J}\}$
$t_s(J)$	Scheduled arrival time at the destination of J
$dest(J)$	Destination of journey J
$Tr(J^i, J^j) \in \{0, 1\}$	A transfer between leg J^i and leg J^j, which is 1 when the transfer is reached and 0 if it is missed
$Pr(Tr)$	Probability to reach transfer Tr given no previous alternative was missed, predicted by transfer model
$Pr(Tr \mid Tr_{prev})$	Probability to reach transfer Tr given that previous alternative transfer T_{prev} was missed, predicted by transfer model
$alternative(J, destination)$	Alternative journey to $destination$ given the last leg of J was missed
$p(Delay(J^i))$	Distribution of delays for the arrival of leg J^i predicted by the delay model

Table 3
Features used in the transfer model.

Feature	Definition
PTT	Planned time between the arrival of the incoming and the departure of the outgoing leg
prevPTTDiff	Difference between the PTT of this transfer and the PTT of a previous transfer if available, NA otherwise
Weekend	1 on a weekend, 0 otherwise
arr.IntercityHour	Interaction variable, if arriving train is an intercity train then the variable represents an integer value equal to the current hour (24h clock), 0 otherwise
arr.ShortTrain	1 if the total runtime of the arriving train is under 2 h, 0 otherwise
arr.IntercityWinter	Interaction variable, 1 if it is winter (December–February) and the arriving train is an intercity train, 0 otherwise
dep.IntercityTrain	1 if the departing train is an intercity train, 0 otherwise

the incoming train of 5 h, the model predicted a 90% transfer reliability. But just changing the runtime to 4.9 h, while leaving all other features the same, resulted in a prediction of 71% transfer reliability. This behavior is most likely due to clusters in the training data at certain runtime values and not due to any real and explainable patterns.

Due to similar issues with the PTT, a monotone constraint was enforced on this feature in the XGBoost model definition. This step ensures that a higher PTT always results in an equal or higher predicted transfer reliability.

Model evaluation. Imbalanced classification requires special consideration when selecting the evaluation metrics. Accuracy is not a suitable metric, as it only evaluates the predicted class labels, and even a baseline model that always predicts transfers to be reached would achieve a close to perfect 95% accuracy. Instead, the probability of reaching or missing a transfer is the central model output used later to calculate the reliability of a journey, and therefore to be evaluated. AUROC (Area Under the Receiver Operator Characteristic) plots the true positive rate against the false positive rate and scores the area under the resulting curve (Ling et al., 2003). A baseline model, choosing randomly between the classes, has an AUROC of 0.5. A perfect model has an AUROC score of 1. Given a random positive and negative sample, the AUROC score can be interpreted as the rate of correctly predicting a higher probability for the positive sample. While the AUROC metric can be used on imbalanced classification problems, it does not directly evaluate the goodness of the predicted probability labels (Lobo et al., 2008). To ensure proper calibration of these labels and the real observed probabilities, we use a calibration curve visualization, which compares the predicted probability labels with the actual observed probabilities. If the predicted probabilities are calibrated well, they follow the diagonal line closely. This visualization is also known as a reliability diagram (Niculescu-Mizil and Caruana, 2005b). Calibration curves and AUROC scores are computed for both the regular test dataset and the external test dataset to evaluate the generalization performance of the model.

4.2. Delay model

The *delay model* gives a probabilistic estimate of the arrival delay $Delay(J^i)$ in the form of a sample of delays following a probability distribution. Marković et al. (2015) notes the difficulty of fitting a single distribution, such as a lognormal or exponential distribution, to train delays. We could confirm this for our data (see Section 5.2) and found that a single distribution could not capture trains with very high delays properly. Therefore, we propose using a probabilistic Bayesian regression model, based on a mixture distribution with two lognormal components, which is better suited to model train delays (Stratil-Sauer, 2024).

A Bayesian model approximates the underlying distribution of the data by treating parameters such as the mean or the variance as probability distributions rather than fixed values. Instead of estimating a single best-fit parameter, it samples from the posterior distribution of these parameters, allowing for a more flexible representation of the variability in train delays. In a Bayesian regression model, the features can be used to model any number of distribution parameters instead of sampling them directly. This is done by including regression coefficients for each feature and sampling a distribution for each coefficient (Gelman et al., 2013).

In our model for train delays, the features are used to model the mean and the variance of both lognormal components of the mixture model:

$$p(x) = \pi_1 \cdot p_1(x \mid \mu_1, \sigma_1) + \pi_2 \cdot p_2(x \mid \mu_2, \sigma_2),$$

where:

$$p_i(x \mid \mu_i, \sigma_i) \sim \text{lognormal}(x \mid \mu_i, \sigma_i),$$

$$\mu_1 = \beta_{1,0} + \beta_{1,1}X_1 + \beta_{1,2}X_2 + \cdots + \beta_{1,k}X_k,$$

$$\mu_2 = \beta_{2,0} + \beta_{2,1}X_1 + \beta_{2,2}X_2 + \cdots + \beta_{2,k}X_k,$$

$$\sigma_1 = \exp(\beta_{3,0} + \beta_{3,1}X_1 + \beta_{3,2}X_2 + \cdots + \beta_{3,k}X_k),$$

$$\sigma_2 = \exp(\beta_{4,0} + \beta_{4,1}X_1 + \beta_{4,2}X_2 + \cdots + \beta_{4,k}X_k).$$

Using Markov Chain Monte Carlo (MCMC) sampling and the R package brms (Bürkner, 2017), coefficients β and mixing weights π are sampled. As this is a Bayesian model, we need to define priors. All priors are set to weakly informative $N(0, 1)$ priors, which have a regularizing effect, resembling ridge regression in a frequentist context (Lemoine, 2019).

The main benefit of the Bayesian regression model for this problem is its probabilistic nature. Its output takes the form of samples following a probability distribution giving detailed characteristics of the uncertainty of the delay. Many other machine learning algorithms would produce point estimates of the predicted delay only. This benefit is especially relevant in the present context of long-term predictions, where the uncertainty is still high, making point estimates less meaningful.

Features. Features used in the model have to be encoded numerically, and since the model is based on linear regression, all non-linear terms and interactions have to be modeled explicitly. The final accepted model uses the features shown in Table 4. As Bayesian regression models can struggle with convergence when faced with a large sample size and many features, the chosen features in the final delay model

Table 4
Features used in the delay model.

Feature	Definition
TotalRuntime	Runtime of the train from its origin to its destination (in hours)
MidDayAfternoonIntercity	Interaction variable, 1 if the time is between 9:00 and 18:00 and the arriving train is an intercity train, 0 otherwise
EveningNightIntercity	Interaction variable, 1 if the time is between 18:00 and 05:00 and the arriving train is an intercity train, 0 otherwise

were carefully selected, which is why they appear more limited than in the transfer model. Features left out in the delay model include the day of the week and the season. Experiments with a smaller sample size showed no significant uplift in model performance when these features were included. The features included act as proxies for the underlying root causes of the delays.

Model evaluation. In Bayesian regression models using MCMC sampling, it is important to ensure convergence of the final model. The metrics $\hat{R}$ and effective sample size ESS are used to guarantee this. Depending on the problem, Bayesian models can struggle to achieve convergence, especially when presented with many features and a big sample size. If these metrics show convergence of the model, it can be accepted (Gelman et al., 2020). Accepted models can be compared according to their expected log-pointwise predictive density (ELPD) on the test dataset (Vehtari et al., 2017). In addition to this, posterior predictive checks can be used to judge the calibration of the model to actual data. If the model is a good fit, it should be able to generate data that cannot easily be distinguished from actual train delay data. We can generate data from the posterior predictive distribution and compare it with actual delays using QQ plots and similar tools (Gabry et al., 2019). These checks can also be performed on the external test data to determine the generalization performance of the model and ensure that it can also reliably model train delays for stations that are not present in the training data.

4.3. Reliability of a journey

In order to predict the reliability of a planned journey J_{plan}, we need to combine the transfer and delay models. Algorithm 1 presents the structure of how both models are combined to produce a delay sample of a journey, including potential transfers. At first, we iterate through the transfers in order to sample an actual journey J_{act}. For each transfer, the transfer model is used to predict the probability of reaching it. This probability is then used to sample whether the transfer was reached or not. If it was not reached, we replace the current journey J with an alternative journey starting at the station where the transfer was missed.

The probability of reaching a transfer is handled conditionally if a previous transfer at the same station was missed through the feature $prevPTTDiff$ in the transfer model. It is set to the difference between the transfer time of the previous transfer and the current transfer time. It is set to NA, if there is no information on previous transfers. This feature represents the fact that a traveler would only consider an alternative transfer if they missed their preferred transfer. Assume that a journey has a transfer with a transfer time of 10 min at station X. In this case, $prevPTTDiff = NA$. But if this transfer is missed, there is an alternative with a transfer time of 30 min. Here $prevPTTDiff = 30 - 10 = 20$, as the knowledge about the previous missed transfer is available. This structure can be seen as a series of binary choices, as shown in Fig. 5. It is important to include this variable because it adds relevant context to the transfer model. If a transfer with a 10-minute transfer time is missed, it is unlikely that an alternative transfer with a 12-minute transfer time would be reached. Using $prevPTTDiff$, this information is considered, and prediction performance improved.

Algorithm 1 Sample a delay $DelaySample$ given a planned journey J_{plan}

Input: Journey J
Output: A delay sample $DelaySample$

$J \leftarrow J_{plan}$
$J_{act} \leftarrow [J^1_{plan}]$
$Tr_{prev} \leftarrow NA$
while $dest(J^{\overline{J}_{act}}_{act}) \neq dest(J^{\overline{J}})$ **do** ▷ Check if we are at the final destination
  $Tr_{next} \leftarrow Tr(J^{\overline{J_{act}}}, J^{\overline{J_{act}}+1})$
  if $Tr_{prev} = NA$ **then**
    $P_{next} \leftarrow Pr(Tr_{next})$ ▷ using the transfer model
  else
    $P_{next} \leftarrow Pr(Tr_{next} | Tr_{prev} = 0)$ ▷ using the transfer model
  end if
  Draw $r \sim \text{Bernoulli}(P_{next})$
  if $r = 1$ **then** ▷ Transfer reached, add to journey
    $J_{act} \leftarrow J_{act} \cap [J^{\overline{J_{act}}+1}]$
    $Tr_{prev} \leftarrow NA$
  else ▷ Transfer missed, update planned journey with alternative
    $Tr_{prev} \leftarrow Tr_{next}$
    $J \leftarrow alternative(J_{act} \cap [J^{\overline{J_{act}}+1}], dest(J))$
  end if
end while
Draw $FinalDelay \sim p(Delay(J^{\overline{J}_{act}}_{act}))$ ▷ using the delay model
$DelaySample \leftarrow t_s(J_{act}) - t_s(J_{plan}) + FinalDelay$
return $DelaySample$

The process is repeated until arrival at the final station, where it results in a final delay and works with any number of transfers and alternative trains. At this point, we have sampled an actual journey J_{act} which is based on both the probability of missing a transfer and the consequences for the onward journey. However, the arrival delay at the destination does not only depend on the trains reached but also on the possible delay of the train used to reach the destination. For this purpose, we use the delay model and sample an arrival delay for the final train used.

The arrival delay sample finally returned by the algorithm is the sum of $t_s(J_{act}) - t_s(J)$ and the final delay sampled by the delay model. The first term accounts for the fact that arriving with a later leg because of a missed transfer adds delay to the journey compared to the planned journey. If all transfers are reached as planned, then this term is 0. The second term accounts for the delay of the last train used. Fig. 5 illustrates this process and how a $DelaySample$ is generated for a journey with one transfer. In this example, the first transfer is sampled to be missed and therefore an alternative transfer has to be used.

Details on how to obtain alternative journeys $alternative(J, destination)$ in case of a missed transfer are not in the scope of this paper. A simple approach is to use the next train connecting the same stations as the missed leg. Alternatively, a public transport routing API could be used, allowing for more flexible re-routing options. In practice, there has to be a cut-off when a journey cannot reasonably be completed. This could be after, for example, 3 missed transfers in a single station. In such a case, the algorithm would terminate and designate that delay as NA, indicating that the final station was not reached.

Model evaluation. To obtain an approximation of an arrival delay distribution for a given journey, the process is executed multiple times to obtain a representative sample of delays. This sample can be compared to actual observations of the journey using the actual arrival and departure data of the involved trains. To evaluate the combined model

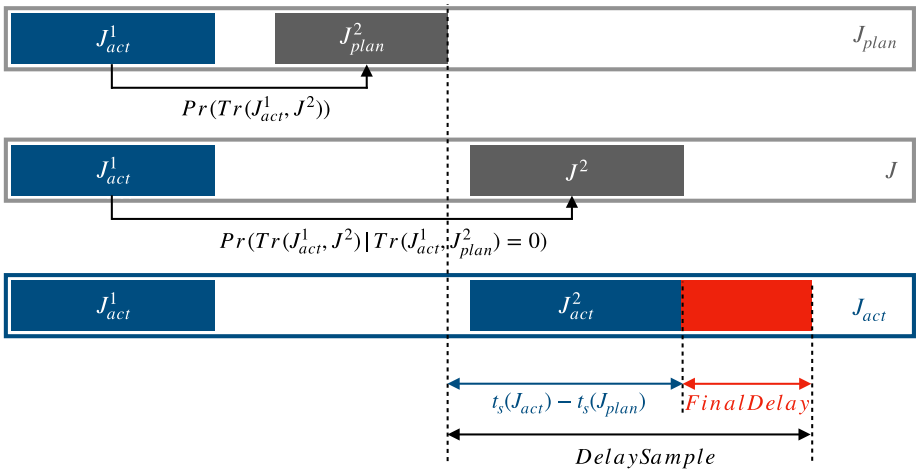

Fig. 5. Example of the sample generating process for a planned journey J_{plan} with one transfer. In the first two steps, the transfer model is used to sample the actual journey J_{act} shown in blue. The planned transfer is missed, and an alternative train for the second leg has to be taken. Finally, the delay model is used to sample a final delay for the last train used, shown in red. Together, the output of both models yields one $DelaySample$.

Table 5
Optimal XGBoost hyperparameters after CV grid search.

Hyperparameters	Value
nrounds	500
max_depth	5
eta	0.1
subsample	0.8
gamma	0.1

structure, delays from real journeys can be compared with sampled delays. Journeys are evaluated by comparing them with real-world examples of the same journey. Evaluation metrics are the reliability rating, which describes the rate of journeys where all transfers were reached, and the reliability buffer time, which measures the variability in travel time, based on the difference between the 95th and the 50th percentile travel time (see Section 2 for more details on these two metrics), as well as through qualitative checks of the resulting density plots.

5. Results

We trained the models using the data described in Section 3. In this section, we present the results, including the validation of the models. At first, the transfer and delay models are evaluated one by one, and finally, the output of the combined journey reliability prediction method is compared with delays from actual train journeys.

5.1. Transfer model

After the hyperparameter optimization, we obtain the optimal hyperparameter configuration as shown in Table 5.

The most important feature for the final model by the gain of predictive performance is the $prevPTTDiff$, followed by the PTT and $arr.hourIC$ features. The AUROC score on the regular test dataset is 0.943, while on the external test dataset, the score is 0.905. Fig. 6 shows the calibration curves of the model on the two test datasets.

The AUROC scores indicate a good prediction performance. Scores are similar between the regular and external test datasets, indicating a good generalization performance on unseen data and stations that the model was not trained on. The calibration curves show that the predicted probability values resemble the true probability quite closely. In the case of the external test dataset, the model appears to be slightly too optimistic for transfers with a low probability of being missed.

Table 6
Summary of the regression coefficients for the delay model. Note that parameters for σ are on the log scale.

Parameter	Mean	l-95% CI	u-95% CI
π_1	0.28	0.28	0.29
π_2	0.72	0.71	0.72
μ_1 *Intercept*	2.03	2.01	2.05
σ_1 *Intercept*	−0.45	−0.47	−0.42
μ_2 *Intercept*	1.79	1.79	1.80
σ_2 *Intercept*	−1.64	−1.65	−1.62
μ_1 *TotalRuntime*	0.15	0.14	0.16
μ_1 *MidDayAfternoonIntercity*	0.03	−0.01	0.07
μ_1 *EveningNightIntercity1*	0.47	0.42	0.52
σ_1 *TotalRuntime*	0.04	0.03	0.05
σ_1 *MidDayAfternoonIntercity*	0.35	0.32	0.37
σ_1 *EveningNightIntercity*	0.26	0.22	0.29
μ_2 *TotalRuntime*	0.01	0.00	0.01
μ_2 *MidDayAfternoonIntercity*	0.01	0.00	0.02
μ_2 *EveningNightIntercity*	0.12	0.10	0.13
σ_2 *TotalRuntime*	0.13	0.12	0.13
σ_2 *MidDayAfternoonIntercity*	0.19	0.17	0.21
σ_2 *EveningNightIntercity*	0.35	0.32	0.38

5.2. Delay model

In the Bayesian modeling workflow (Gelman et al., 2020), we first fit a number of models with different numbers of features, check them for convergence, and then choose the preferred model based on the ELPD score. The resulting mixture model includes one component with a lower mixture weight π_1, but a higher variance σ_1 so that high delays are also given some probability mass. The other component has a higher mixture weight π_2 and lower variance σ_2 and is mainly responsible for representing trains with a lower or no delay (see Table 6).

To check the fit of the final model output compared with actual train delays, we can refer to posterior predictive checks in the form of QQ plots. The output of a well-calibrated model should follow the diagonal line, while deviations indicate problems. Fig. 7 presents this for the regular and external test datasets. While the fit for the test dataset appears to follow the diagonal line almost perfectly, a slight deviation is apparent for the external test dataset. This concave deviation indicates that the sampled delays are slightly more left-skewed than the actual delays of the external test data.

Fig. 8 shows the QQ plot for a model with just one lognormal component for comparison. Here we find significant deviations from the diagonal line, showing the same trend for both test datasets. This

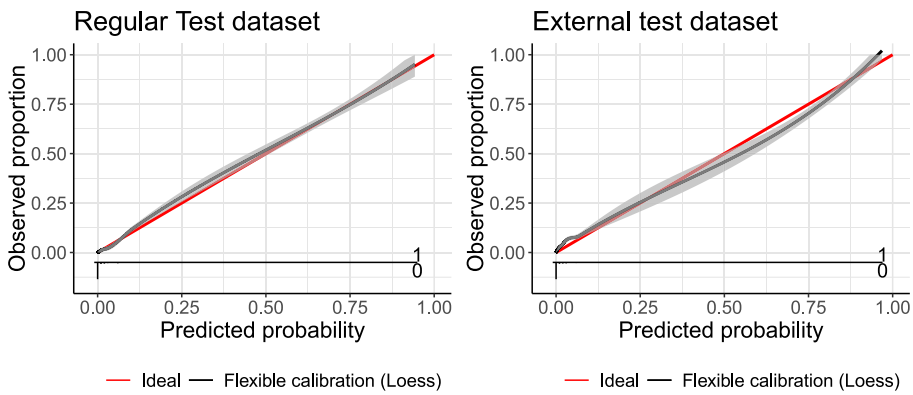

Fig. 6. Comparison of calibration curves of the transfer model with the actual probabilities from the test dataset on the left and the external test dataset on the right. In this case, the model predicts the probability of missing a transfer; lower values indicate higher transfer reliability.

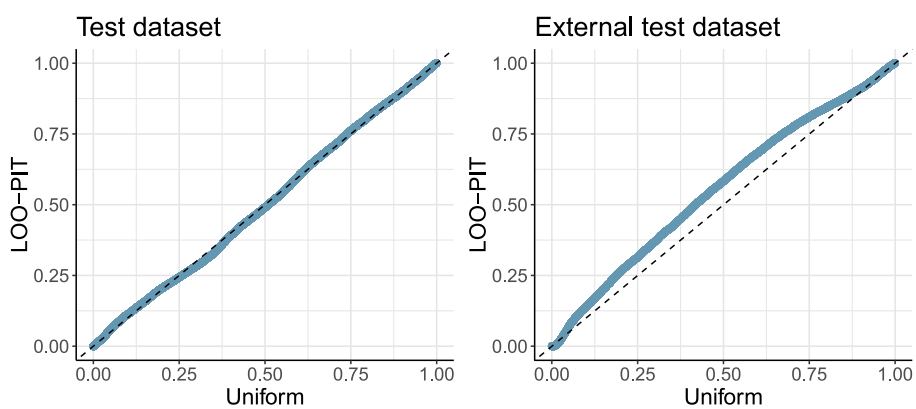

Fig. 7. QQ plot comparison of the posterior predictive output of the delay model with the test and external test datasets.

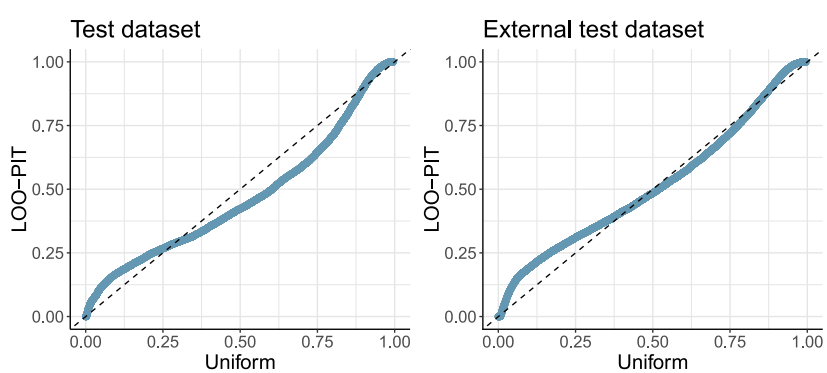

Fig. 8. QQ plot of a lognormal model with just one component used on the regular test dataset.

indicates that this model is not well calibrated. The S curve shows a systematic bias, where the model underestimates the uncertainty in one part of the distribution and overestimates it in the other.

5.3. Evaluating entire journeys

In order to validate the reliability predicted for whole journeys using the method presented in Section 4.3, we selected ten actual journeys (see Appendix). To obtain the initial planned itinerary and to calculate alternative routes in case of a missed transfer, we use the API of the Swedish public transport journey planner Resrobot.[3] For each day in the dataset, we get an actual delay based on the transfers reached that day and the observed delay of the last train. We compare this actual sample to a predicted sample of 1000 delays obtained by repeatedly executing Algorithm 1.

Fig. 9 shows the predicted delay distribution compared with the actual distribution of delays for Journey 1. Each peak in the distribution represents one arriving train at the final station. The first peak is the cases where all transfers have been reached, and the second peak is cases where a transfer was missed and the next train, which arrives 60 min later, was taken.

Fig. 10 shows a comparison of the reliability rating and the reliability buffer time obtained from the actual sample and the sample predicted by our method for all ten journeys. The reliability rating shows how likely it is to reach the destination of the journey without

[3] https://www.trafiklab.se/api/our-apis/resrobot-v21/

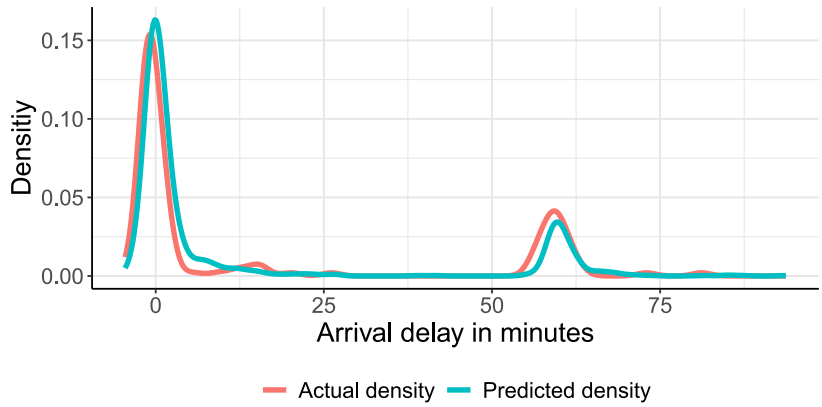

Fig. 9. Comparison of the actual distribution of delays of Journey 1 with the sampled distribution of delays generated by the model.

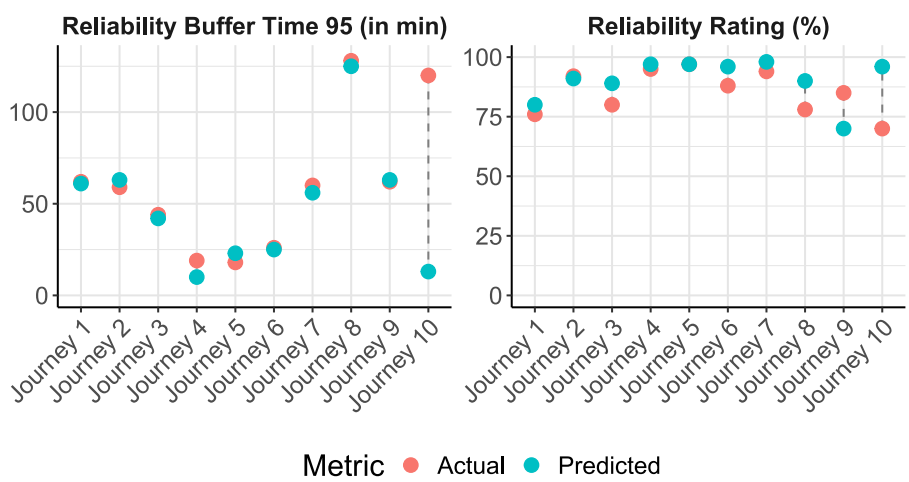

Fig. 10. Comparison of the actual reliability of 10 example journeys with the predicted output of the model.

missing a transfer. This metric only takes the transfer model into account, since final delays are irrelevant. The reliability buffer rating gives the difference between the 95th percentile travel time and the 50th percentile time. Here, the entire model structure is validated, since final delays as well as correctly predicting transfers is important.

In most cases, both metrics are close to the actual observed values. For journeys 8, 9, and 10, on the other hand, the error is bigger, especially considering the reliability rating. Journey 10 shows a big difference between the actual and the predicted reliability buffer time. Possible reasons behind these discrepancies for these journeys are discussed in Section 6.

6. Discussion

The results show that the proposed model structure can be used to predict the distribution of delays and the reliability of transfers for a given train journey. While both models show good results individually, due to the makeup of the dataset and the combination of a regular classification model and a Bayesian probabilistic model, evaluating the results of the entire model structure is more difficult. We have used ten example journeys. One reason is that the existing dataset is limited to a couple of stations in Sweden. Also, often trains get a slightly modified schedule in the yearly timetable process. This makes it difficult to find stable journeys over time with enough observations.

As the final output is a distribution of delays, the selection of evaluation metrics is not straightforward. For this reason, we select evaluation metrics that are relevant to the eventual user of the model output. Both the reliability rating and the reliability buffer time carry meaningful information for a traveler and can be used to validate model outputs and compare them with actual samples.

Investigating these outcomes for the example journeys shows that the model predictions are close to the actual values in most cases, but there is a significant deviation for Journey 10. This journey includes a transfer from an intercity train to a regional train with a 20-minute transfer time during the afternoon. Comparable transfers in the dataset have 90% reliability. Yet, for Journey 10, the reliability is only 68%. This indicates a problem that seems to be specific to this departure and, therefore, is hard for the model to account for.

As shown in Fig. 10, the transfer model is slightly too optimistic in most cases when it comes to the reliability rating. Boosting methods like XGBoost can be prone to miscalibrated prediction outputs, where probabilities are pushed towards 0 or 1 (Niculescu-Mizil and Caruana, 2005a). Even though this effect is small, it is also visible in the calibration curve for the external test dataset in Fig. 6, where the model is slightly too optimistic for transfers that are likely to be reached. Implementing calibration methods like Isotonic Regression did not significantly help with this issue.

The proposed method does make some simplifications. One of these is the use of a fixed minimum transfer time, which is set to 3 min. This is a simplification since the exact time required to move between trains depends on the station and the exact platforms the trains arrive at, as well as the walking speed of the traveler. The former could be addressed by using a dynamic minimum transfer time depending on the tracks used by the trains and incorporating such information into the transfer model. To analyze the sensitivity of the transfer model with respect to the minimum transfer time, we have trained alternative

Table A.1
Specification of the example journeys used for model evaluation.

Journey	Train	Origin	Destination	N
Journey 1	Intercity Train Regional Train Regional Train	Stockholm (10:21) Alvesta (13:39) Emmaboda (14:34)	Alvesta (13:26) Emmaboda (14:28) Karlskrona (15:16)	157
Journey 2	Regional Train Regional Train Intercity Train	Karlskrona (12:41) Emmaboda (13:32) Alvesta (14:30)	Emmaboda (13:25) Alvesta (14:20) Norrköping (16:20)	126
Journey 3	Intercity Train Regional Train	Stockholm (16:14) Alvesta (19:38)	Alvesta (19:26) Växjö (19:48)	100
Journey 4	Regional Train Regional Train	Sala (13:03) Linköping (16:04)	Linköping (15:49) Mjölby (16:24)	276
Journey 5	Regional Train Regional Train	Älmhult (11:27) Hässleholm (12:13)	Hässleholm (12:02) Karlskrona (14:12)	168
Journey 6	Regional Train Regional Train	Karlskrona (09:49) Hässleholm (11:55)	Hässleholm (11:45) Älmhult (12:18)	127
Journey 7	Regional Train Intercity Train	Värnamo (18:46) Alvesta (19:28)	Alvesta (19:18) Linköping (20:51)	190
Journey 8	Intercity Train Regional Train	Hässleholm (18:47) Alvesta (19:36)	Alvesta (19:26) Värnamo (20:06)	66
Journey 9	Intercity Train Regional Train Regional Train	Borlänge (9:55) Sala (11:04) Norrköping (13:48)	Sala (10:57) Norrköping (13:23) Kimstad (13:58)	79
Journey 10	Regional Train Intercity Train Regional Train	Mjölby (10:25) Linköping (11:04) Alvesta (12:46)	Linköping (10:44) Alvesta (12:26) Rydaholm (12:57)	126

models assuming a minimum transfer time of 2 min and 5 min. The original and alternative models all showed similar performance with AUROC scores between 0.94 and 0.95 on the test dataset.

Network effects and interactions between trains are not fully considered by the model structure used. Delays of one train could indicate problems in the entire network, for example, during a weather event. This interaction is not represented in the model structure, as both models are independent of each other. The transfer model also does not explicitly consider the possibility of two trains waiting for each other. Adding such interactions and network effects could further improve the performance, but would require a much more complex model structure, such as a neural network. Modeling network effects was also limited by the available traffic information data, which, for example, lacks data on vehicle circulations that could explain some delay propagation.

7. Conclusions

We proposed a method for modeling the delays and the reliability of whole train journeys with any number of transfers. It accounts for all relevant factors influencing the arrival delay of a traveler, including the possibility of missing a transfer and accounting consequences caused by this, as well as the delay of the train used to reach the destination. The method is flexible and allows, for example, to use of different ways of finding alternative routes in case of a missed transfer.

Train delays and transfer reliability are modeled accurately overall, even though the model for transfer reliability is slightly too optimistic in some cases. The combined model also shows good performance on stations that are not part of the training dataset, indicating a capability for generalization, which is an important prerequisite for real-world usage.

Our model predicts the long-term reliability of a journey, which could be helpful for travelers during the booking and planning stage of their trip by incorporating the information into a journey planner (Wunderlich et al., 2017). Train operators and planners could also use it to estimate the reliability of transfers and entire journeys during the timetable planning process.

Future research is needed to extend the presented model structure for the short-term prediction of reliability by extending the method to take real-time information on delays into account. This could help travelers during their trip, giving accurate information about the best alternative connection and their expected arrival time in case of a delay. Including additional data sources such as delay causes, demand data, and vehicle circulation data would allow modeling and understanding the underlying causes affecting reliability, and could extend the use of the method to timetable planning. In the real-time case, the uncertainty of the delay is lower, which means that other, more advanced machine learning models, providing point estimates, could be employed instead of the current structure of the delay model.

Another important area to investigate is how the model could be expanded for multi-modal journeys, including other public transit modes in addition to trains. While the model structure is also applicable for these journeys, the individual models would have to be expanded, because the delay distributions would likely differ significantly between modes of transport. Reliability may affect both travel mode and route choice, and an important research direction is thus to investigate how the proposed methods could be applied to improve transport models.

CRediT authorship contribution statement

Nikolaus Stratil-Sauer: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Nils Breyer:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Nils Breyer reports financial support was provided by Swedish Transport Administration (TRV 2023/75678). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The work was supported by the Swedish Transport Administration, grant no. TRV 2023/75678.

Appendix. Detailed example journey information

See Table A.1.

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